Project 11: Installation of Apache Tomcat

Objective

The goal of this project is to install and configure Apache Tomcat 9.0.95 on an Ubuntu system. This involves downloading the Tomcat archive, extracting it to the /opt directory, configuring the server port and user access, and finally starting the Tomcat server to verify that it is running correctly via the browser.

Steps

Step 1: Open Terminal and Update System

- **sudo -i** → Switches the user to root (administrator) mode with full system privileges.
- **sudo apt update** → Updates the package list and upgrades all installed packages automatically without asking for confirmation.

```
vboxuser@Ubuntu:~/Desktop$ cd ..
vboxuser@Ubuntu:~$ sudo -i
[sudo] password for vboxuser:
root@Ubuntu:~# sudo apt update
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 2: Download Apache Tomcat

- **cd /tmp** → Changes the current directory to the system's temporary folder.
- **wget** https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz → Downloads Apache Tomcat 9.0.95 compressed file from the official Apache archive to the system.

```
root@Ubuntu:~# cd /tmp
root@Ubuntu:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-
9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-14 15:02:51--  https://archive.apache.org/dist/tomcat/tomcat-
9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189,
2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.95.tar.gz'

apache-tomcat-9.0.95.tar.gz    100%[===================>]  12.13M  80.2KB/s
in 2m 14s
```

```
2026-04-14 15:05:05 (92.8 KB/s) - 'apache-tomcat-9.0.95.tar.gz' saved
[12715996/12715996]
```

Step 3: Create Directory for Tomcat Installation

- **cd ..** → Moves one level up from the current directory.
- **cd /opt** → Navigates to the /opt directory used for installing optional software.
- **mkdir tomcat** → Creates a new directory named tomcat to store the Tomcat files.
- **ls** → Lists directory contents to verify that the tomcat folder is created successfully.

```
root@Ubuntu:/tmp# cd ..
root@Ubuntu:/# cd opt
root@Ubuntu:/opt# ls
google
root@Ubuntu:/opt# mkdir tomcat
root@Ubuntu:/opt# ls
google  tomcat
```

Step 4: Extract Apache Tomcat Files

- **sudo tar -xvzf** apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1 → Extracts the downloaded Tomcat archive into /opt/tomcat directory by removing the top-level folder and placing all files directly inside it.

```
root@Ubuntu:/# cd /tmp
root@Ubuntu:/tmp# sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --
strip-components=1
apache-tomcat-9.0.95/conf/
apache-tomcat-9.0.95/conf/catalina.policy
apache-tomcat-9.0.95/conf/catalina.properties
apache-tomcat-9.0.95/conf/context.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xml
apache-tomcat-9.0.95/conf/server.xml
apache-tomcat-9.0.95/conf/tomcat-users.xml
apache-tomcat-9.0.95/bin/
apache-tomcat-9.0.95/webapps/
... (files extracted)
```

After extraction, verifying the directory:

```
root@Ubuntu:/opt# cd tomcat
root@Ubuntu:/opt/tomcat# ls
bin  BUILDING.txt  conf  CONTRIBUTING.md  lib  LICENSE  logs  NOTICE  README.md
RELEASE-NOTES  RUNNING.txt  temp  webapps  work
```

Step 5: Configure Tomcat Server (server.xml)

- **cd /opt/tomcat/conf** → Navigates to the Tomcat configuration directory.
- **ls** → Lists configuration files inside the directory.
- **gedit server.xml** → Opens the main Tomcat configuration file for editing.
- **Port Change (8080 → 9090)** → Modifies the default Tomcat port to 9090 to avoid conflicts with other services (Jenkins runs on 8080).

```
root@Ubuntu:/opt/tomcat# cd conf
root@Ubuntu:/opt/tomcat/conf# ls
```

```
catalina.policy      context.xml            jaspic-providers.xsd  server.xml
catalina.properties  jaspic-providers.xml   logging.properties    tomcat-
users.xml  web.xml
root@Ubuntu:/opt/tomcat/conf# gedit server.xml
```

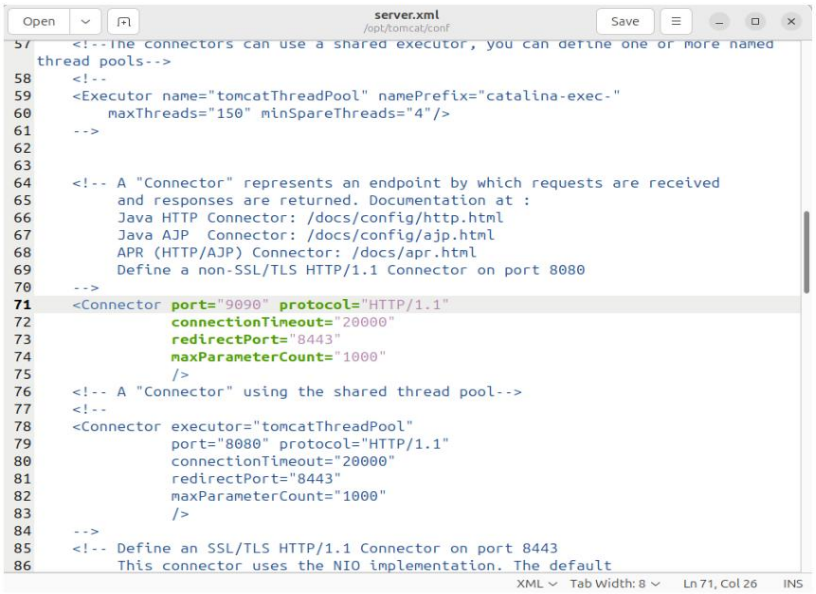

Fig 1: server.xml showing port changed from 8080 to 9090

Step 6: Configure Tomcat User Access (tomcat-users.xml)

- **gedit tomcat-users.xml** → Opens the Tomcat user configuration file.
- **Add user with role** → Creates a user (admin) with password and assigns manager-gui role to access the Tomcat Manager dashboard.

```
root@Ubuntu:/opt/tomcat/conf# gedit tomcat-users.xml
```

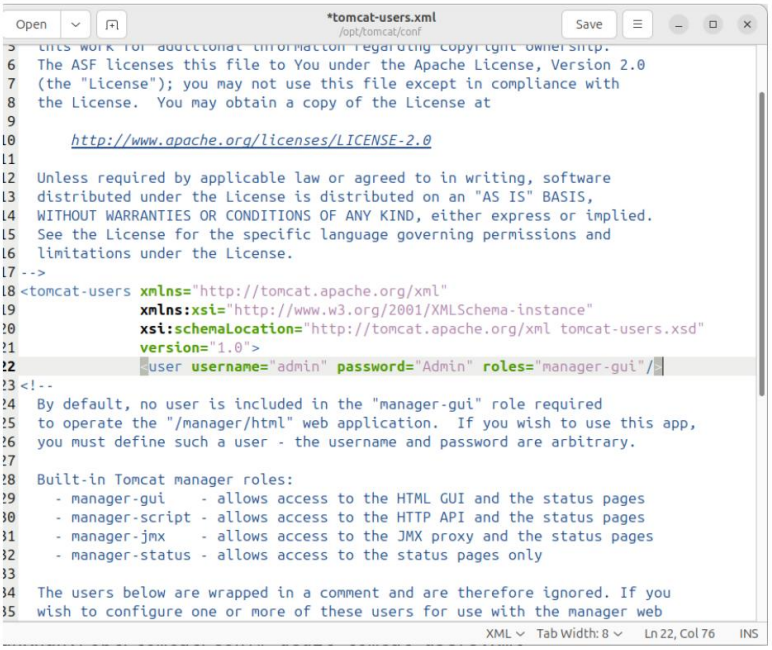

Fig 2: tomcat-users.xml showing admin user with manager-gui role added

Step 7: Start Tomcat Server

- **cd /opt/tomcat/bin** → Navigates to the Tomcat binary directory.
- **ls** → Lists all available startup and shutdown scripts.
- **sh startup.sh** → Starts the Apache Tomcat server successfully.

```
root@Ubuntu:/opt/tomcat# cd bin
root@Ubuntu:/opt/tomcat/bin# ls
bootstrap.jar      ciphers.sh             daemon.sh      setclasspath.sh
startup.sh
catalina.bat       commons-daemon.jar     digest.bat     shutdown.bat
tomcat-juli.jar
catalina.sh        configtest.sh          makebase.sh    shutdown.sh
tomcat-native.tar.gz
root@Ubuntu:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-
juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

Step 8: Verify Tomcat Installation

- **Open Browser (http://localhost:9090)** → Confirms Tomcat server is running successfully by displaying the default Tomcat welcome page.
- **Access Manager (http://localhost:9090/manager/html)** → Logs into the Tomcat Manager dashboard to manage and deploy web applications.

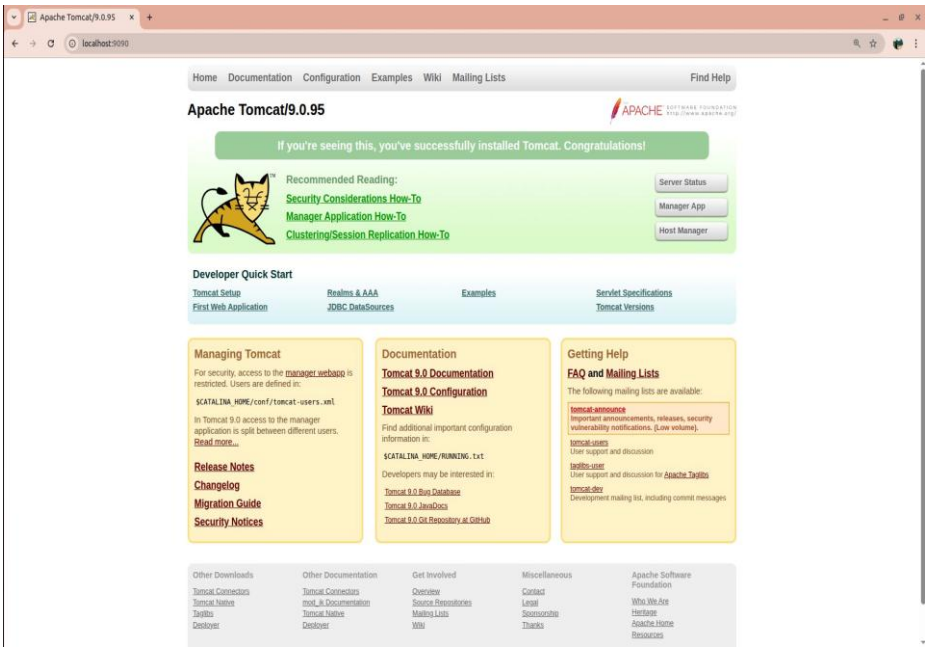

Fig 3: Apache Tomcat 9.0.95 default page confirming successful installation at localhost:9090